Smart Finder and Recovery Management System

Diploma in Computer Engineering (MSBTE)

Matoshri Aasarabai Institute of Technology and Research Centre, Eklahare, Nashik

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Submitted by: Manoj Ambadas Gholap (23611810229), Neel Swapnil Burkule (23611810216), Arpit Dilip Gaikwad (23611810225), Kunal Mahendrasing Solunke (23611810278)

Introduction

A Digital Solution for Recovery

The Smart Finder and Recovery Management System is an innovative web-based platform designed to streamline the process of reporting, searching, and recovering lost items and missing persons. By leveraging modern web technologies, this system bridges the gap between those who lose belongings and those who find them.

Built on the robust Django Framework with MongoDB as its database backbone, the platform ensures efficiency, transparency, and accessibility for all users. This project represents a significant step toward automating traditional lost-and-found operations through technology.

Problem Statement

Manual Processes

Traditional lost-and-found systems rely heavily on manual record-keeping and physical bulletin boards, making them time-consuming and inefficient for both reporters and searchers.

No Centralized Platform

Without a unified digital system, coordination between people who lose items and those who find them becomes fragmented, resulting in poor communication and delayed recoveries.

Verification Challenges

Existing systems lack proper authentication mechanisms, leading to concerns about data integrity, fraudulent claims, and the absence of administrative oversight.

Limited Reach

Geographic constraints and lack of awareness significantly reduce the chances of successful recovery, leaving many items unclaimed and cases unresolved indefinitely.

Project Objectives

01

Connect Users Effectively

Create an intuitive online platform that seamlessly connects individuals who have lost items or persons with those who have found them, facilitating faster resolution.

02

Implement Secure Systems

Deploy robust user registration, authentication, and smart search functionalities that enable quick matching while maintaining user privacy and data security.

03

Ensure Data Integrity

Establish comprehensive administrative controls and verification processes to maintain accurate records and prevent fraudulent activity within the system.

04

Build for Scale

Develop a scalable, easily deployable solution that can grow with user demand while making a meaningful social impact in communities.

System Architecture

Three-Layer Architecture

Presentation Layer

HTML5, CSS3, JavaScript, and Bootstrap create a responsive, user-friendly interface accessible across all devices.

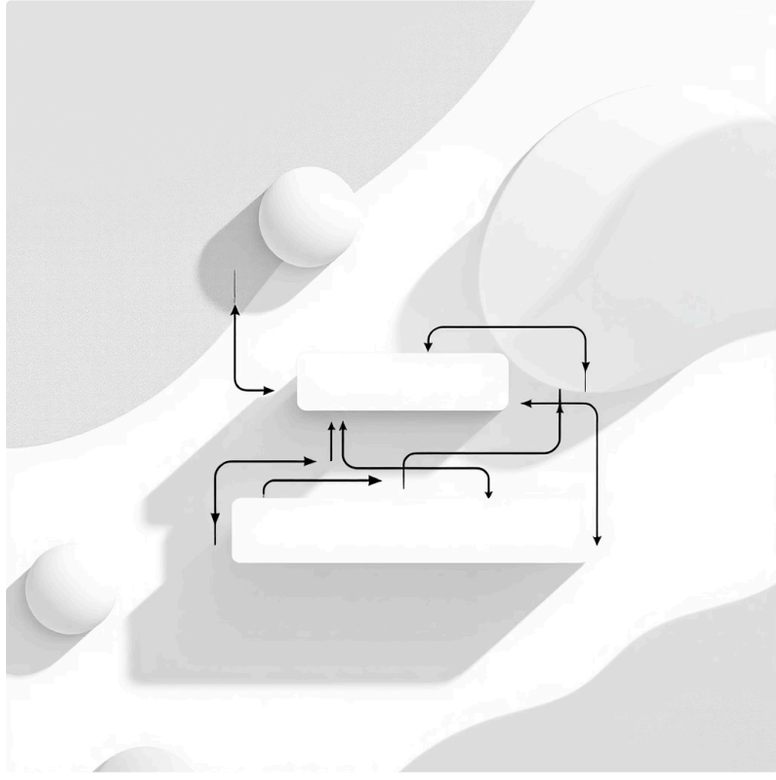
Application Layer

Django Framework handles business logic, request processing, authentication, and data flow management efficiently.

Data Layer

MongoDB provides flexible, scalable NoSQL storage for user data, reports, and matched cases with fast query performance.

Data Flow: User interacts with web interface → Django server processes request → MongoDB stores/retrieves data → System generates response and sends notifications to matched users.



Core Features and Modules

User Registration & Login

Secure authentication system with email verification ensures only verified users can access platform features and submit reports.

Report Lost/Found Cases

Comprehensive reporting interface allows users to submit detailed descriptions, upload images, and specify location information for lost or found items.

Smart Search & Matching

Intelligent algorithm automatically matches lost and found reports based on attributes like category, description, location, and time period.

Admin Verification

Administrative dashboard enables record management, user verification, and oversight to maintain system integrity and prevent misuse.

Alert Notifications

Automated notification system alerts users when potential matches are found, enabling quick communication and faster recovery.

Responsive Design

Mobile-optimized interface ensures seamless experience across desktop computers, tablets, and smartphones for maximum accessibility.

Technology Stack

Frontend Technologies

- HTML5 for semantic structure
- CSS3 for styling and animations
- JavaScript for interactivity
- Bootstrap for responsive layouts

Backend Framework

- Django Framework (Python)
- MVC architecture pattern
- Built-in security features
- ORM for database operations

Database System

- MongoDB (NoSQL)
- Flexible document structure
- Horizontal scalability
- High-performance queries

Development Tools

- Visual Studio Code IDE
- Git for version control
- GitHub for collaboration

Deployment Platform

- Django Development Server
- Render Cloud Hosting
- Scalable infrastructure

Expected Outcomes

Centralized Portal

A unified web platform accessible 24/7 that consolidates all lost-and-found activities in one secure, searchable location.

Faster Recovery

Significantly reduced time between reporting and recovery through automated matching algorithms and instant notifications.

Enhanced Trust

Improved transparency and community participation through verified user accounts and administrative oversight mechanisms.

Future Ready

Solid foundation for integrating advanced features like AI recognition, mobile apps, and government portal connectivity.

Limitations and Future Scope

Current Limitations

Internet Dependency

System requires continuous internet connectivity for access and functionality, limiting use in offline scenarios.

Basic Matching

Current algorithm uses text-based keyword matching without advanced pattern recognition or semantic understanding.

No Real-Time Tracking

Lacks GPS integration and live location tracking capabilities for immediate geographical coordination.

Future Enhancements

AI Image Recognition

Implement machine learning models for automated visual identification and matching of items and persons from uploaded photos.

GPS Integration

Add real-time location services to pinpoint exact recovery locations and enable proximity-based search features.

Mobile Application

Develop native iOS and Android apps with push notifications for on-the-go access and instant alerts.

Government Integration

Connect with police databases and official missing persons registries for comprehensive coverage and legal support.

Conclusion

> ### Leveraging Technology for Social Good
>
> The Smart Finder and Recovery Management System demonstrates how modern web technologies can effectively address real-world challenges. By combining Django's robust framework with MongoDB's flexible data architecture, we've created a scalable, secure, and efficient platform that empowers communities.

Key Takeaways

- Digital transformation of traditional lost-and-found processes
- Enhanced civic responsibility through technology
- Foundation for future AI and mobile innovations
- Scalable solution with significant social impact

References

- Django Documentation (djangoproject.com)
- MongoDB Documentation (mongodb.com)
- W3Schools Web Tutorials
- IJERT & IRJET Research Papers
- CodeWithHarry Python Tutorials
- MSBTE Project Guidelines 2025-26

Thank You!